

Data, Data Types & Sampling – MCQ Quiz (Dr. Mohammed Khalaf)

Instructions

Choose the best answer for each question.

1. What is data?

- a) A random collection of numbers
 - b) Processed information only
 - c) Raw facts gathered for analysis
 - d) A statistical model
-

2. Which of the following is an example of qualitative data?

- a) Age
 - b) Height
 - c) Gender
 - d) Income
-

3. Quantitative data can be divided into:

- a) Primary and Secondary
 - b) Structured and Unstructured
 - c) Discrete and Continuous
 - d) Nominal and Ordinal
-

4. Which of the following is an example of discrete data?

- a) Weight of students
 - b) Height of a building
 - c) Number of students in a class
 - d) Temperature
-

5. Which of the following is continuous data?

- a) Number of books
 - b) Exam grades (A, B, C)
 - c) Speed of a car
 - d) Number of employees
-

6. Structured data is typically stored in:

- a) Images
 - b) Tables and databases
 - c) Audio recordings
 - d) Videos
-

7. Unstructured data includes:

- a) Excel sheets
 - b) SQL databases
 - c) Text documents and images
 - d) CSV files
-

8. The first step in the data collection process is:

- a) Data visualization
 - b) Data cleaning
 - c) Define the objective
 - d) Interpret results
-

9. Which of the following is a source of primary data?

- a) Government reports
 - b) Research articles
 - c) Surveys conducted by the researcher
 - d) Published statistics
-

10. Secondary data is:

- a) Data collected firsthand
 - b) Data already collected by others
 - c) Always qualitative
 - d) Always quantitative
-

11. A population refers to:

- a) A small subset of data
 - b) A random group of participants
 - c) The entire group being studied
 - d) Only survey respondents
-

12. A sample is:

- a) The whole dataset
 - b) A subset of the population
 - c) A statistical error
 - d) A type of qualitative data
-

13. Why do researchers use samples instead of studying the entire population?

- a) To increase bias
 - b) To reduce accuracy
 - c) To save time and cost
 - d) To avoid analysis
-

14. Probability sampling means:

- a) Selecting participants based on convenience
 - b) Every member has a known chance of selection
 - c) Only experts are selected
 - d) Participants volunteer
-

15. Simple random sampling is a type of:

- a) Non-probability sampling
 - b) Qualitative analysis
 - c) Probability sampling
 - d) Secondary data collection
-

16. Stratified sampling is used when:

- a) The population has distinct subgroups
 - b) Data is unstructured
 - c) Only qualitative data is needed
 - d) Time is unlimited
-

17. Convenience sampling is an example of:

- a) Probability sampling
 - b) Random sampling
 - c) Non-probability sampling
 - d) Stratified sampling
-

18. Sampling bias occurs when:

- a) The sample represents the population well
 - b) Some members of the population are more likely to be selected
 - c) The dataset is too large
 - d) The data is continuous
-

19. Which of the following reduces sampling bias?

- a) Selecting friends only
 - b) Using random selection
 - c) Choosing the easiest participants
 - d) Ignoring certain groups
-

20. The main danger of sampling bias is that it:

- a) Increases computation speed
- b) Improves prediction accuracy
- c) Leads to misleading conclusions
- d) Reduces sample size

Answer Key

Q	Answer	Q	Answer
1	c	11	c
2	c	12	b
3	c	13	c
4	c	14	b
5	c	15	c
6	b	16	a
7	c	17	c
8	c	18	b
9	c	19	b
10	b	20	c
